



STEM SMART Single Maths 20 - Data Collection & Binomial Hypothesis Tests >

Sampling and Representations of Data 1

Essential GCSE Maths 54.1

Subject & topics: Maths | Statistics | Data Analysis **Stage & difficulty:** GCSE C1, A Level P1

A hospital has a total of 9760 employees. They are divided into 4 categories:

Doctors	Nurses	Scientists	Administrators
1240	5020	1080	?

Part A

Number of administrators

How many administrators work for the hospital?

Part B

Number of employees surveyed

A stratified sample is required for a survey of employees' opinions. 5% of the employees will be surveyed.

How many employees will be included in the survey?

Part C

Numbers of people from each category

A stratified sample is required for a survey of employees' opinions. 5% of the employees will be surveyed.

Work out how many people must be randomly selected from each of the four employee categories to give a stratified sample.

Number of doctors:

Number of nurses:

Number of scientists:

Number of administrators:

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Woodland Survey

Subject & topics: Maths | Statistics | Data Analysis **Stage & difficulty:** A Level P1

A conservationist wishes to collect data on the number of trees in a woodland nature reserve that have been parasitised by mistletoe. The reserve consists of birch, aspen and oak trees.

Part A

Conducting a census

Why would it be impractical to conduct a census to collect data from every tree in the woodland?

☐ Collecting this data might spread the mistletoe

☐ A sampling frame would be needed.

☐ The act of collecting data from a tree would destroy it.

☐ It would be very time-consuming and expensive.

☐ The population would need to be classified into distinct strata.

Part B

Type of sample

The conservationist wishes to determine if any particular species of tree is more likely to be parasitised by mistletoe. To do so, they decide to survey the first 10 birches, 10 aspen and 10 oak trees that they come across in the reserve.

What type of sample are they collecting?

Part C**Disadvantage of sampling**

What is a disadvantage of the type of sampling described in Part B?

- ☐ It is very time-consuming and expensive.
- ☐ This non-random sampling could introduce bias.
- ☐ A sampling frame is needed in order to select the sample.

Part D**Species data type**

What type of data is the species of a tree?

Part E**Number of trees data type**

What type of data is the number of trees that have been parasitised by mistletoe?

- ☐ Continuous qualitative data.
- ☐ Discrete quantitative data.
- ☐ Discrete qualitative data.
- ☐ Continuous quantitative data.

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Researcher Survey

Subject & topics: Maths | Statistics | Data Analysis **Stage & difficulty:** A Level P1

A research institute wishes to collect data from its employees on the methods by which they travel to work. The institute has an alphabetised list of its 450 employees.

Part A

Systematic sample

Describe how they could take a systematic sample of size 30.

Assign numbers from 1 to to the employees on the alphabetised list. Calculate $450 \div \text{} = \text{}$. Generate a random number between 1 and . Select the employee corresponding to that number and then select every th employee on the list after that.

Items:

450

15

30

Part B

Alternative method

One researcher suggests that it would be much easier to collect a sample by speaking to the first 30 employees in the canteen.

What type of sampling would this be?

Give one disadvantage of this type of sampling.

Part C

Stratified sample

The research institute also wishes to gather data from its research staff as to which new pieces of lab equipment will be required over the next year. The research staff within the institute consist of 40 geneticists, 25 ecologists and 55 epidemiologists. It is thought that the different types of staff will have different requirements for lab equipment.

Describe how the institute could collect a stratified sample of size 20 from its research staff.

There are a total of 120 research staff. The sample is $\div$ 120 = of the research staff.

The institute should survey $40 \times$ $\approx$ geneticists, $25 \times$ $\approx$ ecologists and $55 \times$ $\approx$ epidemiologists. The staff should be selected randomly from each group, by generating random numbers from 1 to for geneticists, 1 to for ecologists and 1 to for epidemiologists and selecting the corresponding members of staff, discarding and replacing any duplicate numbers within each group.

Items:

- 9
- $\frac{1}{6}$
- 40
- 6
- $\frac{5}{24}$
- 25
- 7
- $\frac{1}{3}$
- 5
- 55
- 20
- $\frac{11}{24}$
- 10
- 4

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Hypothesis Testing: Meeples

Subject & topics: Maths | Statistics | Hypothesis Tests Stage & difficulty: A Level P2

A manufacturer of meeples (a type of playing piece used in many board games) finds that, on average, 10% of the meeples they produce are defective. In order to improve the production of their meeples, they purchase new equipment to reduce the number of defects. In order to test the new equipment, a member of their quality assurance team collects a random sample of 80 meeples produced using the new equipment and finds that 3 of them have defects. Test at the 5% significance level whether the new equipment has reduced the number of defects.

Let the test statistic X be the number of meeples with defects. Then $X \sim B(\text{ }, p)$.

The null and alternative hypotheses are:

$H_0 : p \text{ } \text{ } \text{ } H_1 : p \text{ } \text{ } \text{ }$

We find that $P(X \leq \text{ }) = \text{ }$.

Comparing this to the significance level, we find that $P(X \leq \text{ }) \text{ } \text{ } \text{ }$.

Therefore we $\text{ } H_0$ at the 5% level. There evidence to suggest that the new equipment has reduced the number of meeples with defects.

Items:

>

<

=

≠

reject

do not reject

is

is insufficient

0.1

0.05

0.025

3

5

8

10

70

72

80

88

90

0.0107

0.0246

0.0353

0.0527

0.0880

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Hypothesis Testing 3.1

Subject & topics: Maths | Statistics | Hypothesis Tests

Stage & difficulty: A Level P2

An experiment has two possible outcomes, A and B. An experimenter investigates the probabilities with which the two outcomes occur by carrying out the experiment a large number of times; X , the number of times the experimenter gets outcome A, is recorded. Answer the following questions.

Part A

Outcomes equally likely

In a previous set of measurements the two outcomes of the experiment (A and B) have been found to be equally likely. An experimenter wishes to test, at the 5% significance level, the null hypothesis that the two outcomes are equally likely against the alternative hypothesis that the two outcomes are not equally likely. The experiment is carried out 50 times and X , the number of times the experimenter gets outcome A, is recorded.

Find the lower bound X_l of the acceptance region.

$X_l =$

Find the upper bound X_u of the acceptance region.

$X_u =$

Part B

Slight difference in probability of outcomes

In a previous set of measurements the two outcomes of the experiment (A and B) have been found to be equally likely. The experiment is carried out 50 times and the experimenter gets outcome A 34 times i.e. $X = 34$. By answering the following **three** questions use the data to test, at the 1% significance level, the hypothesis that the outcomes are not equally likely.

Assuming that the null hypothesis is correct, find P_p , the probability that $X = 34$. Give your answer to 2 sf.

$$P_p = \text{[]}$$

Assuming that the null hypothesis is correct, find P_c , the probability that $X \geq 34$. Give your answer to 2 sf.

$$P_c = \text{[]}$$

You were asked to use the data above to test, at the 1% significance level, the hypothesis that the outcomes are not equally likely. Using the appropriate answer from earlier is there sufficient evidence to reject, at this level, the null hypothesis that the outcomes are equally likely? Which of the following arguments is correct in this regard?

- ☐ $P_p < 0.005$ so reject the null hypothesis
- ☐ $P_p < 0.01$ so reject the null hypothesis
- ☐ $P_c < 0.01$ so reject the null hypothesis
- ☐ $P_c < 0.005$ so reject the null hypothesis
- ☐ $P_p > 0.01$ so there is not sufficient evidence to reject the null hypothesis
- ☐ $P_c > 0.005$ so there is not sufficient evidence to reject the null hypothesis
- ☐ $P_c > 0.01$ so there is not sufficient evidence to reject the null hypothesis
- ☐ $P_p > 0.005$ so there is not sufficient evidence to reject the null hypothesis

Part C

Probability of A greater than that of B

In a previous set of measurements the two outcomes of the experiment (A and B) have been found to be equally likely. As described in Part B the experiment is carried out 50 times and the experimenter gets outcome A 34 times i.e. $X = 34$. Test, at the 1% significance level, the hypothesis that the probability of getting the outcome A is greater than that of getting the outcome B.

Is there sufficient evidence to reject, at this level, the null hypothesis that the outcomes are equally likely? Which of the following arguments is correct in this regard?

- ☐ $P_p < 0.01$ so reject the null hypothesis
- ☐ $P_c < 0.01$ so reject the null hypothesis
- ☐ $P_p < 0.005$ so reject the null hypothesis
- ☐ $P_c < 0.005$ so reject the null hypothesis
- ☐ $P_p > 0.005$ so there is not sufficient evidence to reject the null hypothesis
- ☐ $P_c > 0.005$ so there is not sufficient evidence to reject the null hypothesis
- ☐ $P_p > 0.01$ so there is not sufficient evidence to reject the null hypothesis
- ☐ $P_c > 0.01$ so there is not sufficient evidence to reject the null hypothesis

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Hypothesis Testing 3.2

Subject & topics: Maths | Statistics | Hypothesis Tests

Stage & difficulty: A Level P2

An experimenter uses a piece of test equipment to make measurements of samples she has manufactured. There is a 10% probability that the equipment will fail when she makes a measurement. She makes some changes to the equipment which she hopes will increase its reliability. Answer the following questions about how she will test whether her changes have increased the reliability.

Part A

Carrying out 30 measurements

The experimenter would like to test the hypothesis, at the 2% level of significance, that her changes have increased the reliability of her equipment. She decides to make 30 measurements and there are no failures.

Assuming that the failure rate is unchanged find the probability that there are no failures. Give your answer to 2 sf.

Deduce what the experimenter can conclude about the changes she has made.

- ☐ There is insufficient evidence at the 2% level to reject the null hypothesis that the changes have had no effect; there is insufficient evidence to indicate that the changes have improved the reliability of the test equipment.
- ☐ Given the original failure rate she would have expected about 3 failures, so there is sufficient evidence to support her hypothesis
- ☐ The changes have not improved the reliability of the test equipment.
- ☐ There is sufficient evidence at the 2% level to reject the null hypothesis that the changes have had no effect; at the 2% level there is evidence to support the hypothesis that the changes have improved the reliability of the test equipment.
- ☐ There is insufficient evidence at the 2% level to reject the null hypothesis that the changes have had no effect; therefore the changes have not improved the reliability of the test equipment.

Part B**Carrying out more measurements**

The experimenter realises that she needs more evidence to test, at the 2% level of significance, whether or not her changes have increased the reliability of her equipment. She decides to make more measurements. Assuming that she still gets no failures, how many measurements will she have to make in total to provide support for her hypothesis at the 2% level of significance?

Part C**Carrying out 60 measurements**

The experimenter makes 60 measurements in all and gets one failure.

Assuming that the failure rate is unchanged find the probability that she will get one failure or no failures. Give your answer to 2 sf.

Hence deduce what the experimenter can conclude about the changes she has made.

- ☐ There is sufficient evidence at the 2% level to reject the null hypothesis that the changes have had no effect; at the 2% level there is evidence to support the hypothesis that the changes have improved the reliability of the test equipment.
- ☐ There is sufficient evidence at the 2% level to reject the null hypothesis that the changes have had no effect; at the 2% level there is insufficient evidence to support the hypothesis that the changes have improved the reliability of the test equipment
- ☐ The number of failures is less than the 6 expected so the null hypothesis can be rejected; the reliability of the test equipment has therefore improved.
- ☐ There is insufficient evidence at the 2% level to reject the null hypothesis that the changes have had no effect; at the 2% level there is insufficient evidence to support the hypothesis that the changes have improved the reliability of the test equipment.
- ☐ There is insufficient evidence at the 2% level to reject the null hypothesis that the changes have had no effect; at the 2% level there is evidence to support the hypothesis that the changes have improved the reliability of the test equipment.

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Hypothesis Testing 3.3

Subject & topics: Maths | Statistics | Hypothesis Tests

Stage & difficulty: A Level P2

According to one theory the probability that a particular result will be obtained in an experiment is 0.6 whereas according to another theory the probability is 0.7. The experiment is carried out 100 times and the particular result is obtained 76 times.

Part A

Testing $p = 0.6$

Consider the first theory which suggests that $p = 0.6$; assume the null hypothesis is that $p = 0.6$ and the alternative hypothesis is that $p > 0.6$. Find the critical region for the test (in this case if X is the number of times the particular result is obtained the critical region has one part with $X \geq X_h$). If the theory is to be tested at the 2% level of significance, deduce X_h .

 $X_h =$

Part B

Testing $p = 0.7$

Now consider the second theory which suggests that $p = 0.7$; assume the null hypothesis is that $p = 0.7$ and the alternative hypothesis is that $p \neq 0.7$. Find the critical region for the test (in this case if X is the number of times the particular result is obtained the critical region has two parts $X \leq X_l$ and $X \geq X_h$). If the theory is to be tested at the 2% level of significance, deduce X_l and X_h .

 $X_l =$ $X_h =$

Part C

Conclusions from the tests

What can be concluded at the 2% level from the tests?

- ☐ There is sufficient evidence to reject the hypothesis that $p = 0.6$ and to support the theory that $p > 0.6$; there is sufficient evidence to reject the hypothesis that $p = 0.7$ and to support the theory that $p \neq 0.7$.
- ☐ There is sufficient evidence to reject the hypothesis that $p = 0.6$ and to support the theory that $p > 0.6$; the hypothesis that $p = 0.7$ is correct.
- ☐ There is insufficient evidence to reject the hypothesis that $p = 0.6$; there is insufficient evidence to reject the hypothesis that $p = 0.7$.
- ☐ There is insufficient evidence to reject the hypothesis that $p = 0.6$; there is sufficient evidence to reject the hypothesis that $p = 0.7$ and to support the theory that $p \neq 0.7$.
- ☐ There is sufficient evidence to reject the hypothesis that $p = 0.6$ and to support the theory that $p > 0.6$; there is insufficient evidence to reject the hypothesis that $p = 0.7$.
- ☐ The hypotheses that $p > 0.6$ and that $p = 0.7$ are correct.

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